

*MA.912.F.1.2***Benchmark**

MA.912.F.1.2 **Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.**

Benchmark Clarifications:

Clarification 1: Problems include simple functions in two-variables, such as $f(x, y) = 3x - 2y$.

Clarification 2: Within the Algebra I course, functions are limited to one-variable such as $f(x) = 3x$.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.2
- MA.912.AR.3.8
- MA.912.AR.4.3
- MA.912.AR.5.6
- MA.912.F.3.1

Terms from the K-12 Glossary

- Function Notation

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.1.2

Next Benchmarks

- MA.912. F.3

Purpose and Instructional Strategies

In middle grades, students worked with x - y notation and substituted values in expressions and equations. In Algebra I, students work with x - y notation and function notation throughout instruction of linear, quadratic, exponential and absolute value functions. In later courses, students will continue to use function notation with other function types and perform operations that combine functions, including compositions of functions.

- Instruction leads students to understand that $f(x)$ reads as “ f of x ” and represents an output of a function equivalent to that of the variable y in x - y notation.
- Instruction includes a series of functions with random inputs so that students can see the pattern that emerges (*MTR.5.1*).
 - For example,

$$f(x) = 2x^2 + 5x - 7$$

$$f(k) = 2k^2 + 5k - 7$$

$$f(-2) = 2(-2)^2 + 5(-2) - 7$$

- Students should discover that the number in parenthesis corresponds to the input or x -value on the graph and the number to the right of the equal sign corresponds to the output or y -value.
- Although not conventional, instruction includes using function notation flexibly.
 - For example, function notation can be written as $h(x) = 4x + 7$ or $4x + 7 = h(x)$.

- Instruction leads students to consider the practicality that function notation presents to mathematicians. In several contexts, multiple functions can exist that we want to consider simultaneously. If each of these functions is written in x - y notation, it can lead to confusion in discussions.
 - For example, representing the equations, $y = -2x + 4$ and $y = 3x + 7$, in function notation allows mathematicians to distinguish them from each other more easily (i.e., $f(x) = -2x + 4$ and $g(x) = 3x + 7$).
- Function notation also allows for the use of different symbols for the variables, which can add meaning to the function.
 - For example, $h(t) = -16t^2 + 49t + 4$ could be used to represent the height, h , of a ball in feet over time, t , in seconds.
- Function notation allows mathematicians to express the output and input of a function simultaneously.
 - For example, $h(3) = 7$ would represent a ball 7 feet in the air after 3 seconds of elapsed time. This is equivalent to the ordered pair $(3, 7)$ but with the added benefit of knowing which function it is associated with.

Common Misconceptions or Errors

- Throughout students' prior experience, two variables written next to one another indicate they are being multiplied. These changes in function notation and will likely cause confusion for some of your students.
- Students may need additional support in the order of operations.
 - For example, for exponential functions, many students multiply a by the growth factor and then raise the product to the value of the exponent.
 - For example, students may think that multiplication is always performed before division.

Strategies to Support Tiered Instruction

- Instruction includes discussing the meaning of function notation with students with understanding that $f(x)$ does not mean $f \cdot x$.
- Instruction is provided to determine the order of operations required once a given input is placed into the function for evaluation. Students may need additional support determining the correct order of operations to perform.
- Teacher models using parentheses to help organize order of operations when evaluating functions.
 - When evaluating $f(x) = 4x^2$ for $x = -1$, teacher can model the use of parentheses by writing the expression $4(-1)^2$ rather than without using parentheses writing $4 \cdot -1^2$. This will help students visualize the operations.

- Teacher provides instruction for identifying the operations in various functions as they relate to the order of operations using a graphic organizer.

Find $f(5)$	$f(x) = 4x^3 - 3.5x^2 + 10$
Substitute 5 for x .	$f(5) = 4(5)^3 - 3.5(5)^2 + 10$
Evaluate the exponents.	$f(5) = 4(125) - 3.5(25) + 10$
Multiply the factors within each term.	$f(5) = 500 - 87.5 + 10$
Perform addition and subtraction of terms from left to right.	$f(5) = 412.5 + 10$ $f(5) = 422.5$

Instructional Tasks

Instructional Task 1 (MTR.3.1)

The original value of a painting is \$9,000 and the value increases by 7% each year. The value of the painting can be described by the function $V(t) = 9000(1 + 0.07)^t$, where t is the time in years since 1984 and $V(t)$ is the value of the painting.

Part A. Create a table of values that corresponds to this function.

Part B. Graph the function.

Instructional Items

Instructional Item 1

Evaluate $f(24)$, when $f(x) = \frac{3}{2}x + 9$.

Instructional Item 2

Given $f(x) = 2x^4 - 0.24x^2 + 6.17x - 7$, find $f(2)$.

Instructional Item 3

A shipping company charges a base fee of \$20 plus an additional \$5 for each package shipped. The total cost (C) of shipping p packages can be modeled by the function $C(p) = 20 + 5p$ where p is the number of packages. You are a small business owner looking to send out 10 packages to customers. Evaluate the function C for $p = 10$ to find the total cost of shipping the packages.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*